
MIXTURE OF LAYERS WITH HYBRID ATTENTION: PARALLEL THIN BLOCKS FOR SPARSE TRANSFORMER COMPUTE

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Abstract

Standard Mixture-of-Experts (MoE) transformers route tokens to expert subnetworks within each layer, but the layer structure itself remains monolithic. We introduce **Mixture of Layers** (MoL), which replaces full-width transformer blocks (d_{model}) with K parallel *thin blocks* at reduced dimensionality ($d_{\text{thin}} \ll d_{\text{model}}$), connected via learned down/up projections and composed via top- k block routing. Scaling sparse block routing to many blocks creates an attention coverage problem, as each block sees fewer tokens. We address this by introducing **hybrid attention**, which pairs one shared softmax block for global context with Gated DeltaNet [Yang et al., 2024] linear attention in routed blocks.

On WikiText-103 at 85M params, MoL with dense FFN thin blocks reaches PPL 30.95 ± 0.11 (3 seeds), surpassing the rank-1 expressiveness ceiling of traditional MoE by 2.98. Adding hybrid attention (1+3of15, 198M total, 77M active) reaches PPL 29.99 ± 0.08 with up to $4.9\times$ forward-pass speedup from sparse dispatch. On Cosmopedia v2 (15B tokens), MoL overtakes dense baselines past 35% of training and finishes at PPL 6.49 vs 6.65; a dense DeltaNet control at 6.64 indicates the gain is not driven by the attention swap alone.

Scaling to FineWeb-Edu 20B tokens at $d_{\text{model}}=2048$, MoL Hybrid 1+3of15 (2.08B total / 0.61B active) reaches PPL 18.04 in a single-seed run; this is 0.49 PPL better than an iso-active Dense Softmax 0.7B baseline (18.53) but 3.01 PPL worse than iso-total Dense Softmax 1.3B (15.03), and takes $1.91\text{--}2.29\times$ longer to train than each dense baseline due to gradient-checkpointing pressure on the routed-DeltaNet kernel. Zero-shot transfer on an 8-task lm-eval-harness suite places MoL between the two dense baselines as capacity predicts, with a clean win on WinoGrande over Dense Softmax 1.3B (54.4 vs 51.6). In single-GPU prefill, MoL Hybrid crosses Dense Softmax 1.3B at $T \approx 5\text{--}6\text{K}$ on the RTX 3090. On datacenter GPUs (A100, H100 SXM, H200), the crossover occurs between $T=64\text{K}$ and $T=128\text{K}$. At

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$T=128K$, MoL ranges from near-parity ($0.98\times$ on H100 SXM) to $1.20\times$ on A100; at $T=256K$ it reaches $1.42\times - 1.54\times$ on H100 SXM and H200. The four measured points do not support a simple bandwidth- or compute-linear scaling, and we do not claim a hardware scaling law. At 1.3B, MoL trades training efficiency and total-parameter capacity for an iso-active perplexity and downstream-transfer lead, plus a long-context prefill advantage that activates at much shorter context on the 3090 than on the three datacenter GPUs in our setup.

1 Introduction

Scaling language models traditionally means stacking more sequential transformer blocks, which increases depth, memory footprint, and latency in lockstep. Mixture-of-Experts (MoE) transformers [Fedus et al., 2022, Lepikhin et al., 2021, Shazeer et al., 2017, Zoph et al., 2022] address parameter scaling by routing tokens to expert subnetworks, but the layer structure remains monolithic; every position runs a single full-width transformer block.

In a companion paper [Ternovtsii and Bilak, 2026], we found that routing topology is quality-neutral across structurally different MoE variants. If topology is not the primary driver of performance, what is? Prior work points to per-expert expressiveness: 65,536 rank-1 experts cannot replicate 256 rank-256 experts despite matched parameters (PPL 47.51 vs 12.55; see Appendix A).

We investigate whether layer structure itself can be decomposed, presenting four contributions:

1. **Mixture of Layers (MoL).** MoL replaces full-width transformer blocks with K parallel thin blocks at $d_{\text{thin}}=256$, connected via learned down/up projections. With dense FFNs, MoL achieves PPL 30.95 ± 0.11 (3 seeds, 85M params) on WikiText-103, 2.98 better than the full-width MoE baseline, surpassing the rank-1 expressiveness ceiling (Section 2).
2. **Hybrid attention for sparse routing.** Scaling to more blocks degrades attention quality as each block sees fewer tokens. To address this, one shared softmax block (global coverage) is paired with Gated DeltaNet [Yang et al., 2024] routed blocks ($O(T)$ linear attention on sparse subsets). DeltaNet outperforms softmax by 0.85 PPL after controlling for parameters in the thin-block regime ($d_{\text{thin}}=256$), an advantage that vanishes at $d_{\text{thin}}=d_{\text{model}}$ (Section 3).
3. **Sparse dispatch with wall-clock speedup.** Sparse dispatch (gather/scatter) yields up to $4.9\times$ forward-pass speedup at high sparsity. DeltaNet 1+3of15 (198M total, 77M active, 3 seeds) achieves PPL 29.99 ± 0.08 , activating only 27% of blocks per token (Section 2.3).
4. **Regime-dependent scaling.** MoL’s advantage is data-regime dependent: it loses to dense on WikiText-103 (103M tokens, multi-epoch) but wins on Cosmopedia v2 (15B tokens, single epoch), with a 0.85 PPL swing across datasets. A dense DeltaNet control matches dense softmax to within 0.01 PPL, indicating MoL’s gain is not driven by the attention swap (Section 5.3). At 1.3B-scale on FineWeb-Edu 20B tokens (single seed each), MoL Hybrid (2.08B total / 0.61B active, PPL 18.04) sits between iso-active Dense Softmax 0.7B (18.53) and iso-total Dense Softmax 1.3B (15.03): 0.49 PPL ahead at iso-active, 3.01 PPL behind at iso-total, with a 1.91–2.29 $\times$ training-time tax that does not extend to inference (Section 5.5).

Unlike sequential hybrid designs that alternate attention types across layers [Glorioso et al., 2024, Qwen Team, 2025], MoL maintains full attention coverage at every layer through the shared block while confining linear attention to sparse routed subsets.

2 The MoL Architecture

2.1 Thin Block Wrapper

MoL replaces selected full-width transformer blocks with **split stages**, each containing K independent thin blocks. Each thin block consists of a down-projection, a standard transformer block at reduced dimensionality, and an up-projection.

$$\text{ThinBlock}(x) = W_{\text{up}} \cdot (\text{Block}_{d_{\text{thin}}}(W_{\text{down}} \cdot x) - W_{\text{down}} \cdot x) \quad (1)$$

where $W_{\text{down}} \in \mathbb{R}^{d_{\text{thin}} \times d_{\text{model}}}$, $W_{\text{up}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{thin}}}$, and $\text{Block}_{d_{\text{thin}}}$ is a complete transformer block (RMSNorm $\rightarrow$ multi-head attention $\rightarrow$ RMSNorm $\rightarrow$ FFN) operating at dimensionality d_{thin} . The subtraction strips the inner residual, so ThinBlock outputs only the delta computed by the block.

2.2 Split Stage with Block Routing

A split stage runs K thin blocks, selects k via a learned router, and averages their outputs:

$$\text{SplitStage}(x) = x + \frac{1}{k} \sum_{i \in \text{top-}k(K)} w_i \cdot \text{ThinBlock}_i(x) \quad (2)$$

where w_i are softmax-normalized routing scores and top- k selects the k highest-scoring blocks. The outer residual $x + \dots$ ensures gradient flow even if thin blocks produce small updates. Selective block activation outperforms uniform composition by 0.99 ± 0.10 PPL (3 seeds, $\sim 9\times$ pooled seed std $\sigma_{\text{pooled}}=0.12$).

Load balancing. We use a coefficient-of-variation (CV^2) loss on per-block routing weights, weighted by $\alpha=0.05$, following standard MoE practice. This prevents block collapse without requiring auxiliary tokens or capacity constraints.

RoPE compatibility. All configurations maintain $d_{\text{head}}=64$: $d_{\text{model}}=1024$ uses 16 heads, $d_{\text{thin}}=256$ uses 4 heads. Precomputed RoPE embeddings are shared across all block widths.

2.3 Sparse Dispatch

With block-level routing, each thin block processes only its routed tokens. We implement *sparse dispatch*, which **gathers** only routed tokens per block, runs on compact tensors, and **scatters** results back. This eliminates wasted compute on non-routed tokens.

Correctness. Sparse dispatch produces numerically identical outputs to dense restricted attention (max logit diff 5.7×10^{-6} on GPU, 2.4×10^{-7} on CPU).

Speedup. Forward-pass speedup scales with sparsity, reaching $1.53\times$ at 57% active (top-4-of-7), $2.85\times$ at 20% (top-2-of-10), and $4.34\times$ at 10% (top-2-of-20). With `torch.compile`: up to $4.94\times$.

Projection overhead. Projection cost ($W_{\text{down}}, W_{\text{up}}$) sets a floor on d_{thin} : at $d_{\text{thin}}=256$ projections consume 40% of wrapper parameters, rising to 57% at $d_{\text{thin}}=128$ and 73% at $d_{\text{thin}}=64$. Empirically $d_{\text{thin}}=256$ outperforms $d_{\text{thin}}=128$ by 1.62 PPL at iso-parameters ($\sim 85\text{M}$), so width outweighs count.

3 Hybrid Attention

3.1 The Attention Coverage Problem

Scaling to more blocks with sparse dispatch creates an attention coverage problem. At 3-of-15, each block sees only 20% of the sequence (mean inter-token gap of 25.6), compared to 60% at 3-of-5 (gap of 1.8). Softmax-only sparse 3-of-15 achieves PPL 34.73 on WikiText-103, worse than 3-of-5 (32.04) despite using $2.2\times$ more parameters; more parameters cannot compensate for reduced sequence coverage.

3.2 Architecture: Shared + Routed

Each split stage contains:

- **Shared block** (block 0): always active on all tokens, full softmax attention, providing global context at every layer.
- **Routed blocks** (blocks 1– $N-1$): top- k selected by the router, sparse dispatch with Gated DeltaNet [Yang et al., 2024] linear attention on their token subsets.

We denote configurations as $S+K\text{of}N$: S shared blocks plus top- K routed from $N-S$ candidates, with N total blocks. For example, 1+3of15 has 1 shared softmax block and selects 3 from 14 routed DeltaNet blocks, activating 4 total.

3.3 Why DeltaNet in Routed Blocks

Gated DeltaNet is used in routed blocks; since the shared block already supplies global softmax context, the routed blocks do not require precise quadratic attention. A d_{thin} scaling sweep also shows DeltaNet

outperforming softmax in the constrained-width regime ($d_{\text{thin}}=256$), and DeltaNet is $O(T)$ rather than $O(T^2)$, becoming faster at $T \geq 2048$.

Decomposing the DeltaNet advantage. At $K=1$ (single block, no routing, all tokens visible), DeltaNet outperforms softmax by 2.55 PPL (36.28 vs 38.83) at matched architecture ($\sim 43\text{M}$ params). An iso-parameter softmax control (87.5M, $d_{\text{ff}}=1133$, 3 seeds: 32.09 ± 0.26) reveals that DeltaNet’s extra parameters (gates, convolutions) account for 0.26 PPL (23%) of the 1.11 PPL gap in the 1+2of5 MoL configuration; the remaining 0.85 PPL (77%) is the parameter-controlled mechanism contribution, $3.0\times$ the pooled seed std ($\sigma_{\text{pooled}}=0.29$).

Scale dependence. A scaling sweep across $d_{\text{thin}} \in \{128, 256, 512, 1024\}$ shows the DeltaNet–softmax gap shrinks monotonically; the gap is $2.79 \rightarrow 2.55 \rightarrow 1.12 \rightarrow -0.10$ PPL. At $d_{\text{thin}}=d_{\text{model}}=1024$, softmax marginally wins (27.60 vs 27.70). DeltaNet’s inductive biases (decay, delta rule) help most when attention capacity is constrained; the advantage vanishes when capacity is sufficient.

4 Experimental Setup

4.1 Datasets

We evaluate on three datasets of increasing scale and tokens-per-parameter: **WikiText-103** [Merity et al., 2017]: 103M tokens, custom BPE 32K, $T=255$, multi-epoch (~ 8 epochs); architecture ablations at 43–198M params. **Cosmopedia v2** [Ben Allal et al., 2024]: 15B synthetic-textbook tokens, GPT-2 vocab, $T=2048$, single-epoch; data-regime dependence at $\sim 104\text{M}$ params. **FineWeb-Edu** [Penedo et al., 2024]: filtered web tokens, GPT-2 vocab, single-epoch. 80M topology sweep at $T=2048$ ($\sim 2.5\text{B}$ tokens at 38K steps, Exp 079); 1.3B-scale runs at $T=4096$ (20B tokens, Exp 085/087/089).

4.2 Architectures

80M-scale ablations (WikiText-103, Cosmopedia): $d_{\text{model}}=1024$, 16 heads, 8 layers, $d_{\text{thin}}=256$, 4 heads per thin block. Training: 50K steps (WikiText) or 114K steps (Cosmopedia), $\text{lr} = 3 \times 10^{-4}$, cosine decay, 1000-step warmup (WikiText) / 2000-step (Cosmopedia), AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\text{wd}=0.01$), gradient clipping at 1.0, effective batch size 64.

1.3B-scale headline (FineWeb-Edu 20B tokens, $4\times$ H200 NVL). Three architectures share $d_{\text{model}}=2048$, 24 layers, GPT-2 50,257-vocab, $T=4096$, batch 1.0M tok/step, 20K steps, AdamW $\text{lr} = 2 \times 10^{-4}$ cosine to $0.1\times$, 2000-step warmup, bf16 with gradient checkpointing. **Dense Softmax 1.3B** (Exp 085): $d_{\text{ff}}=8192$, 16 heads ($d_{\text{head}}=128$). 1.31B total. **Dense Softmax 0.7B** (Exp 089): $d_{\text{ff}}=2048$ matched to a single MoL thin block, 32 heads ($d_{\text{head}}=64$ matched to thin-block heads). 0.71B total. **MoL Hybrid 1+3of15** (Exp 087): $d_{\text{thin}}=512$, 15 thin blocks (1 shared softmax + 14 routed DeltaNet, top-3 active = 4 active per token), $d_{\text{ff,thin}}=2048$. 2.08B total / 0.61B active.

4.3 Baselines

- **Dense Softmax**: standard transformer, iso-param to MoL conditions. At 85M: PPL 30.26 (WikiText), 6.65 (Cosmopedia). At 198M: PPL 26.89 (WikiText).
- **Dense DeltaNet**: DeltaNet attention, iso-param. At 85M: PPL 30.68 (WikiText). At 104M: PPL 6.64 (Cosmopedia), tying dense softmax within 0.01 and confirming the attention mechanism does not explain MoL’s structural advantage.
- **CT-MoE baseline** [Ternovtsii and Bilak, 2026]: rank-1 MoE, 84.7M params, PPL 33.93 (WikiText).

5 Results

5.1 Breaking the Rank-1 Ceiling with Dense FFN Blocks

Replacing rank-1 MoE FFNs with standard dense FFNs ($d_{\text{ff}} = 4 \times d_{\text{thin}}$) in thin blocks exceeds the rank-1 ceiling (Table 1). Dense FFN MoL ($K=5$, top-3, 85.3M params) achieves PPL 30.95 ± 0.11 (3 seeds), 2.98 better than the full-width rank-1 baseline (33.93).

Table 1: Dense FFN thin blocks vs rank-1 MoE on WikiText-103. All at $d_{\text{thin}}=256$, 50K steps.

Config	FFN	Params	PPL	Δ vs CT-MoE
CT-MoE baseline (rank-1)	MoE	84.7M	33.93	—
MoL K=7 split_all (rank-1)	MoE	85.6M	34.83	+0.90
MoL K=5 top-3 (dense)	Dense	85.3M	30.95 ± 0.11	-2.98
MoL K=5 all-active (dense)	Dense	85.3M	31.94 ± 0.12	-1.99

Table 2: Hybrid attention on WikiText-103. S=shared (all tokens), R=routed (sparse). $\dagger=1$ seed; $\pm=3$ seeds.

Config	Attn	Active	Params	PPL
Dense full-attn K=5 †	softmax	3/5	85.3M	30.85
Sparse 3-of-5	softmax	3/5	85.3M	32.19 ± 0.15
Softmax 1+2of5 iso-param	softmax	1S+2R	87.5M	32.09 ± 0.26
DeltaNet 1+2of5	S:smx, R: Δ	1S+2R	87.5M	31.24 ± 0.31
Softmax 1+2of15 †	softmax	1S+2R	190.4M	36.51
DeltaNet 1+2of15 †	S:smx, R: Δ	1S+2R	198.2M	32.75
DeltaNet 1+3of15	S:smx, R: Δ	1S+3R	198.2M	29.99 ± 0.08
Dense softmax 85M †	softmax	—	84.7M	30.26
Dense softmax 198M †	softmax	—	198.2M	26.89
Dense DeltaNet 85M †	DeltaNet	—	84.7M	30.68

5.2 Hybrid Attention Results

Table 2 summarizes the hybrid attention experiments on WikiText-103. DeltaNet 1+2of5 (31.24 ± 0.31 , 3 seeds) outperforms both iso-param Softmax 1+2of5 (32.09 ± 0.26) and standard Softmax 1+2of5 (32.35 ± 0.15). Scaling to 15 blocks, DeltaNet 1+3of15 (198.2M total, 77M active) achieves PPL **29.99 ± 0.08** (3 seeds), activating only 27% of blocks per token.

Dense baselines contextualize MoL quality. At iso-total-parameters (198M), a dense softmax transformer reaches PPL 26.89, 3.1 PPL better than DeltaNet 1+3of15. This establishes the cost of sparse block routing. Note that MoL activates only 77M parameters per token (39% of total), so this gap is paid against a model with $2.6\times$ fewer per-token FLOPs and admits block-parallel placement across devices.

5.3 Data-Regime Scaling: Cosmopedia 15B

On WikiText-103 (103M tokens, ~ 8 epochs), MoL consistently underperforms dense: MoL K=5 top-3 (85.3M, 30.95 ± 0.11) loses by 0.69 PPL vs dense softmax (30.26). Training on Cosmopedia v2 (15B unique tokens, single epoch) reverses this (Table 3).

MoL K=5 crosses the dense baseline at step 40K (35% through training) and stabilizes at -0.16 PPL, representing a **0.85 PPL swing** relative to WikiText-103. The sparse MoL variant (DeltaNet routed + softmax shared) tracks within 0.03 PPL (6.52 vs 6.49), retaining 81% of the MoL gain under sparse dispatch.

Isolating the structural contribution. Dense DeltaNet (103.4M, Cosmopedia) finishes at PPL 6.64, within 0.01 of dense softmax (6.65). Since both dense baselines converge to the same PPL, MoL’s 0.15–0.16 PPL lead cannot be attributed to the linear attention choice. The advantage derives from block-level routing with joint attention+FFN specialization. An MoE-FFN ablation (shared attention + routed FFN experts, no block routing) finishes at PPL 6.89 (+0.24 vs dense), confirming that FFN-only routing does not explain MoL’s gain.

5.4 FineWeb-Edu 80M Ablations

We test whether the MoL advantage on Cosmopedia (Section 5.3) replicates on web-text by running an 80M-scale sweep on FineWeb-Edu for 60K steps ($T=2048$, ~ 3.9 B tokens, $3\times$ local GPUs, single seed each),

Table 3: Cosmopedia v2 results (15B tokens, single epoch, ~ 104 M params). MoL overtakes dense at step 40K.

Condition	Params	Final PPL	vs Dense
Dense softmax	103.4M	6.65	—
Dense DeltaNet	103.4M	6.64	−0.01
MoL K=5 top-3	104.0M	6.49	−0.16
Sparse MoL 1+2of5	106.0M	6.52	−0.13
MoE-FFN 3of5	~ 106 M	6.89	+0.24

comparing three MoL Hybrid topologies (1+3of{5, 10, 15}) against two Dense Softmax baselines bracketing both ends of the active-parameter spectrum (Table 4).

Table 4: FineWeb-Edu 80M sweep at 60K steps (Exp 079). MoL conditions use shared softmax and routed Gated DeltaNet ($d_{\text{thin}}=256$). Differences are relative to the Dense 98M baseline. Single seed each.

Condition	Total	Active	Val PPL @ 60K	Δ PPL
Dense Softmax 152M (iso-FLOP)	152M	152M	24.97	−4.61
Dense Softmax 98M (iso-active)	97.7M	97.7M	29.58	—
MoL Hybrid 1+3of15	217M	95M	26.40	−3.18
MoL Hybrid 1+3of10	162M	95M	27.76	−1.82
MoL Hybrid 1+3of5	106M	95M	27.96	−1.62

At iso-active compute (Dense 98M with $d_{\text{ff}}=768$ matched to MoL’s ~ 95 M active), all three MoL topologies cross Dense between steps 10K–12K and finish 1.62–3.18 PPL ahead by 60K, with the gap widening through training. **At iso-FLOP** (Dense 152M, $1.6\times$ more active compute), Dense finishes 1.43 PPL ahead of best MoL, confirming that MoL trades total parameters for active-compute efficiency, the same trade as MoE but at the block level. A K_{active} sweep at fixed pool $N=5$ (Section I) shows the result holds across different block counts.

5.5 1.3B-Scale Results

We trained three architectures on FineWeb-Edu 20B tokens to compare MoL Hybrid against Dense Softmax baselines bracketing it on both ends of the active-parameter spectrum.

Table 5: 1.3B-scale validation perplexity on FineWeb-Edu 20B tokens ($4\times$ H200 NVL, identical optimizer / token budget). Wall time is the full 20K-step run.

Model	Total	Active	d_{ff}	Final PPL	Wall time
Dense Softmax 1.3B (Exp 085)	1.31B	1.31B	8192	15.03	53.3h
Dense Softmax 0.7B (Exp 089)	0.71B	0.71B	2048	18.53	44.5h
MoL Hybrid 1+3of15 (Exp 087)	2.08B	0.61B	2048 [†]	18.04	102.0h

[†] Per thin block; 4 active blocks per token.

Iso-active comparison. On the closest active-parameter match, MoL Hybrid (0.61B active) reaches 18.04 PPL vs Dense Softmax 0.7B (0.71B active, d_{ff} matched to a single thin block) at 18.53, a 0.49 PPL gap; MoL crosses Dense 0.7B at step 7K and stabilizes at $\sim 1.03\times$ ratio. The 1.3B comparison uses one seed per condition; cross-seed std on the same architecture is $\sigma \approx 0.08$ PPL on WikiText-103 (Section G).

Iso-total gap. Dense Softmax 1.3B finishes 3.01 PPL ahead of MoL’s 2.08B total; the gap stabilizes at step 11K (Figure 1). Candidate causes (active FFN capacity, within-MoL sparsity, sub-Chinchilla token budget) are discussed in Section J.

Zero-shot transfer. On an 8-task lm-eval-harness suite (7 tasks in Table 6; TruthfulQA mc1 in Section K), MoL wins 4 of 8 tasks against iso-active Dense 0.7B (HellaSwag +1.0, ARC-Challenge +0.6, PIQA +0.1, WinoGrande +5.0) and edges Dense 1.3B on WinoGrande (54.4 vs 51.6). MoL trails Dense 1.3B on the

broad-knowledge tasks (HellaSwag, ARC-Easy, PIQA) by 1.5–4.9 pts, consistent with the iso-total PPL gap, and trails iso-active Dense 0.7B on TruthfulQA mc1/mc2 (a documented inverse-scaling regime). All three sit at chance on MMLU 5-shot ($\sim 24\%$), expected at 1.3B / 20B tokens; the pattern places MoL’s downstream transfer at fixed active compute between the density-matched dense baselines, as capacity predicts.

Table 6: Zero-shot accuracy (%), lm-eval-harness 0.4.7, bf16; **acc_norm** where benchmark prefers it; MMLU 5-shot, others 0-shot). Best per row in bold; MMLU rows are within noise of chance ($\sim 25\%$) so we omit bolding there. Stderr per task is 0.005–0.014.

Task	Dense 1.3B (085)	Dense 0.7B (089)	MoL 0.61B/2.08B (087)
MMLU (5-shot)	24.31	23.98	23.07
HellaSwag	39.71	34.10	35.10
ARC-Easy	53.07	48.86	48.15
ARC-Challenge	26.88	25.43	26.02
PIQA	65.56	63.98	64.09
WinoGrande	51.62	49.33	54.38
TruthfulQA mc2	37.83	40.89	39.55

Training cost. MoL Hybrid takes $1.91\times$ Dense 1.3B’s wall time (102.0h vs 53.3h) at identical token budget. Two costs compound. First, gradient checkpointing is mandatory at 2.08B total params on 141 GB H200s. Second, DeltaNet’s chunked kernel is more expensive to replay during backward than FlashAttention. On Cosmopedia 80M (no checkpointing), the same architecture was $1.45\times$ *faster* than Dense (Section E); the training tax is scale- and memory-budget-specific, not intrinsic.

Inference inversion at long context. The training penalty inverts at inference (single-GPU, batch=1, prefill). On RTX 3090, MoL Hybrid is $1.25\times$ – $1.76\times$ faster than Dense Softmax 1.3B at $T=8$ – 32 K. On the three datacenter GPUs we measured (A100, H100 SXM, H200), the crossover sits between $T=64$ K and $T=128$ K, with MoL ranging from near-parity ($0.98\times$ on H100 SXM) to $1.20\times$ on A100 at $T=128$ K and $1.42\times$ – $1.54\times$ at $T=256$ K (Section 5.6). At $T=256$ K, MoL Hybrid 2.08B successfully completes prefill on A100 80GB whereas Dense Softmax 1.3B OOMs, a memory-pressure manifestation of MoL’s smaller per-layer softmax KV cache. The four measured points do not support a clean bandwidth- or compute-linear scaling story; we report the empirical crossovers and leave a multi-architecture mechanistic model to future work. At 1.3B, MoL trades a training premium and an iso-total PPL gap for an iso-active PPL and downstream-transfer lead, plus a long-context prefill advantage whose onset is hardware-specific.

5.6 Inference Wall-Clock: Prefill and Decode

We benchmark prefill and decode throughput at batch=1 on two GPU classes (bf16, post-warmup, FlashAttention-2 for softmax, FLA Triton kernels for DeltaNet, KV cache for both Dense and MoL): a memory-bandwidth-limited **RTX 3090** (936 GB/s HBM, 71 TFLOPS bf16) and a compute/bandwidth-rich **H200** (4.8 TB/s HBM, 989 TFLOPS bf16). For the 3090 prefill table we add two control architectures to isolate the routing contribution from the attention-type contribution: a *Dense DeltaNet 1.3B* (iso-architecture to Dense Softmax with the attention swapped) and a *MoL all-softmax 2.08B* (same routing as MoL Hybrid but with `routed_attn_type=softmax`).

5.6.1 Prefill: Long Context

Empirical crossover positions across four GPUs. On RTX 3090, MoL Hybrid crosses Dense Softmax at $T\approx 5$ – 6 K and reaches $1.76\times$ Dense at $T=32$ K. On the three datacenter GPUs we measured (A100 80GB, H100 SXM 80GB, H200 144GB), the crossover sits between $T=64$ K and $T=128$ K despite a $3\times$ HBM-bandwidth spread ($1.55\rightarrow 4.8$ TB/s) and a $3\times$ compute spread ($312\rightarrow 989$ TFLOPS bf16): MoL reaches $1.20\times$ Dense at $T=128$ K on A100, $\approx 1.0\times$ on H100 SXM, $1.05\times$ on H200, and $1.42\times$ – $1.54\times$ Dense at $T=256$ K on H100 SXM and H200 (Dense OOMs at $T=256$ K on A100 80GB while MoL Hybrid 2.08B does not, isolating the memory advantage of MoL’s smaller per-layer softmax KV cache). *The four measured points do not support a simple monotonic scaling with HBM bandwidth or compute alone.* Plausible contributing factors include differences in FlashAttention kernel efficiency across architectures, GDDR6X vs HBM memory hierarchies, L2 cache sizes, and SM-to-bandwidth ratios; we do not isolate these and we do not claim a closed-form scaling law. We report the empirical crossovers above as observed in our single-GPU, batch=1, FA-2/FLA bench. Our earlier H200 sweep stopped at 32K and led us to incorrectly conclude no crossover

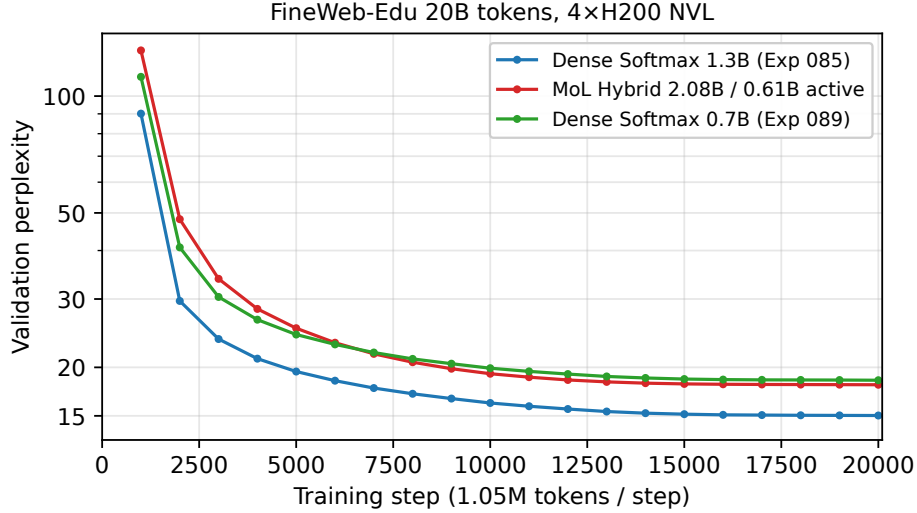


Figure 1: Validation perplexity (log scale) vs training step for the three 1.3B-scale runs on FineWeb-Edu 20B tokens. Dense Softmax 1.3B (blue) leads from step 1; MoL Hybrid (red) stays ~ 3 PPL above through training; Dense Softmax 0.7B (green) starts above MoL and crosses below at step 7K, then narrows the gap from -3.5 PPL to -0.49 PPL by step 20K. The Dense 1.3B vs MoL gap stabilizes at step 11K and does not close within budget.

existed on Hopper; the 4-hardware data shows the crossover is universal in our measurements but late on datacenter GPUs.

Decomposing the 3090 speedup. The per-token latency reduction on 3090 at $T=32K$ (Dense Softmax $92.5 \mu s \rightarrow$ MoL Hybrid $52.5 \mu s$) decomposes into a routing-structure contribution of $31.6 \mu s$ (**79%**, Dense Softmax $\rightarrow$ MoL all-softmax: smaller per-block d_{ff} and 14 narrow routed-attention windows of $\sim T/4.67$ tokens vs one giant $T \times T$) and a DeltaNet-on-routed contribution of $8.4 \mu s$ (**21%**, MoL all-softmax $\rightarrow$ MoL Hybrid: $O(T)$ linear attention on the routed pathway). Dense DeltaNet 1.3B reaches $1.90 \times$ Dense Softmax at $T=32K$ on 3090 (the $O(T)$ vs $O(T^2)$ effect alone). Both decomposition runs are 3090-only; an H200 decomposition would require re-running the controls and is left for future work.

Implementation. The numbers above use the production inference path `attention_mode="batched_sparse"`, which fuses the $N=14$ routed FLA calls into one `chunk_gated_delta_rule` call with $N \times H$ heads and stacks per-block linear projections via `bmm`. The naive (sparse) path is $2.8 \times$ slower than batched on 3090 at $T=256$ batch=1, and $2.77 \times$ slower on H200 at $T=4K$ (consistent across hardware). Numerical equivalence is preserved: validation perplexity on the Exp 087 trained checkpoint shifts only -0.11% ($18.48 \rightarrow 18.46$, within bf16 noise).

Decode (3090 KV-cache). On 3090 with KV cache enabled, Dense Softmax decode latency grows linearly at ~ 3 ms per 4K context (6.2 ms/tok @256 $\rightarrow$ 26.5 ms/tok @24K, $4.3 \times$), while MoL stays flat at ~ 60 – 65 ms across the same range, dominated by a per-token dispatch floor of $\sim 2,160$ Python op launches; the asymptotic flat-vs-linear property holds but the floor makes MoL slower than Dense in absolute terms across all measured contexts. Detailed decode tables and the H200 decode follow-up are deferred to Section H.

Caveats. The Dense DeltaNet 1.3B prefill control uses random weights (Exp 086 was planned but not trained, so we attribute wall-clock without claiming joint quality+speed); at $T \leq 2K$ batch=1 on 3090, Dense Softmax still wins by up to $0.78 \times$ on prefill.

5.7 Limitations

Four limitations scope our claims. **(1)** 1.3B uses single-seed runs; cross-seed $\sigma \approx 0.08$ PPL on 80M WikiText-103 (Section G) does not bound scale variance. **(2)** MoL’s 1.91 – $2.29 \times$ training cost at 1.3B reflects gradient-checkpointing pressure on the routed-DeltaNet kernel; at 80M Cosmopedia (no checkpointing) the same ar-

Table 7: Prefill throughput (tokens/s, batch=1, single GPU; 5-run mean per cell after warmup, run-to-run variance <2% at fixed T on each GPU). MoL Hybrid value is the faster of **batched_sparse** and **sparse** at each T (both shown where measured; see protocol note). *Best per row in bold* marks the crossover between Dense and MoL Hybrid. The four measured GPUs do not show a clean monotonic shift in crossover position with HBM bandwidth or compute; we present empirical positions only.

SeqLen	Dense Softmax 1.3B	Dense DeltaNet 1.3B [†]	MoL all-softmax 2.08B [†]	MoL Hybrid 2.08B
<i>RTX 3090 (consumer; 936 GB/s HBM, 71 TFLOPS bf16)</i>				
4,096	18,041	17,962	20,591	16,005
32,768	10,808	20,502	16,424	19,062
<i>A100 SXM4 80GB (1.55 TB/s HBM, 312 TFLOPS bf16)</i>				
4,096	43,585	—	—	9,998
32,768	28,472	—	—	16,650
65,536	19,495	—	—	15,968
131,072	11,779	—	—	14,115 (sparse)
<i>H100 SXM 80GB (3.35 TB/s HBM, 989 TFLOPS bf16)</i>				
4,096	98,476	—	—	16,459
32,768	71,925	—	—	33,579
65,536	51,731	—	—	33,559
131,072	31,299	—	—	30,819 (sparse, $\approx 0.98\times$ Dense)
262,144	17,495	—	—	24,841 (sparse)
<i>H200 (4.8 TB/s HBM, 989 TFLOPS bf16)</i>				
4,096	102,460	—	—	14,174
32,768	71,912	—	—	36,911
65,536	51,860	—	—	36,292 (sparse) / 35,244 (batched)
131,072	31,348	—	—	33,008 (sparse)
262,144	16,973	—	—	26,105 (sparse)

[†] Random-initialized control architectures (not trained); used on 3090 to attribute the speedup to routing vs attention-type. **Path-selection protocol:** **batched_sparse** stacks all $N=14$ routed FLA calls into one kernel (peak ~ 85 GB at 128K) and dominates short context where dispatch overhead matters; **sparse** dispatches per block (peak ~ 32 GB at 128K) and dominates long context where the stacked footprint OOMs. We report the faster path at each T . **Bench limits:** Dense Softmax 1.3B OOMs above 524K (H200) and 128K (A100/H100 SXM 80GB); MoL sparse OOMs above 262K with the current KV-cache implementation; FlashAttention-3, fp8 KV-cache, or multi-GPU serving could move all ceilings.

chitecture was $1.45\times$ faster than Dense, so the tax is memory-budget-specific. **(3)** Prefill advantage is single-GPU, batch=1 on FlashAttention-2 + FLA kernels, with hardware-specific onset ($T\approx 5\text{--}6$ K on RTX 3090; $T=64\text{--}128$ K on A100/H100/H200; Section 5.6). Multi-GPU parallelism, batched serving, FlashAttention-3, and fp8 KV-cache could shift the crossover; we report a controlled microbenchmark, not a deployment result. MoL decode is slower at every measured context ($3.1\text{--}9.7\times$) and we do not extrapolate beyond 24K. **(4)** The 1.3B-scale comparison uses FineWeb-Edu 20B tokens at ~ 15.6 tokens/active-param; larger budgets and additional corpora are needed to test whether the iso-total gap persists.

6 Related Work

MoE and layer modularity. Standard MoE [Fedus et al., 2022, Shazeer et al., 2017, Zoph et al., 2022] routes tokens to FFN experts within a full-width layer. In contrast, MoL routes over whole transformer blocks at reduced d_{model} . Layer-level modularity work modifies *which* layers execute (MoD [Raposo et al., 2024], MoLEx [Teo and Nguyen, 2025], MoR [Bae et al., 2025], MoEUT [Csordás et al., 2024]); sub-network parallelism partitions weights (UoE [Yang et al., 2025]), the FFN only (FlashMHF [Zhang et al., 2026]), or sub-blocks (AltUp [Baykal et al., 2023]); MoM [Gong et al., 2024] and MoUE [Chen et al., 2026] pool/share modules, all at full d_{model} . No prior work combines parallel blocks, reduced- d_{model} down/up projections (used as fine-tuning in adapters/LoRA [Houlsby et al., 2019, Hu et al., 2022]), top- k block routing, and hybrid attention.

Hybrid attention. Unlike sequential hybrids that alternate attention types across layers [Glorioso et al., 2024, Gu and Dao, 2024, Peng et al., 2023, Qwen Team, 2025], MoL carries softmax (shared) and $O(T)$ DeltaNet [Yang et al., 2024] (routed) *within* every layer.

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A The Rank-1 Expressiveness Ceiling

Before decomposing layer structure, we established the fundamental quality constraint: per-expert expressiveness. A rank-1 mirror experiment compares 65,536 rank-1 experts against 256 rank-256 experts at matched total expert parameters ($\sim 67\text{M}/\text{layer}$) and matched active compute ($K=1024$ rank-1 vs $K=4$ rank-256).

Table 8: Rank-1 mirror experiment. 65,536 rank-1 experts cannot replicate 256 rank-256 experts.

Model	N/layer	Rank	Params	PPL
High-rank reference	256	256	$\sim 290\text{M}$	12.55
Rank-1 mirror	65,536	1	322.7M	47.51
CT-MoE Wide 1×12	1,024	1	84.7M	33.93

The rank-1 mirror achieves PPL 47.51, $3.8\times$ worse, despite 11% more total parameters. This establishes that the rank-1 constraint is the binding quality limit: architectural rearrangements operate within this ceiling, motivating the transition to dense FFN thin blocks (Section 2).

B Active Compute Scaling

Scaling from $K=12$ to $K=128$ active rank-1 experts ($10\times$ more active FLOPs, $N=4096$, 138M params, $\tau=15$) yields PPL 33.22, closing only 19% of the 3.67-PPL gap to the dense baseline (30.26) despite $10\times$ more active computation. This confirms that quality is limited by *per-expert expressiveness*, not the number of active experts.

C Activation-Level Gating

Squeeze-and-Excitation (SE) gating [Hu et al., 2018] applied to expert activations (not d_{model} -dimensional vectors) achieves PPL 31.82, the best iso-parameter rank-1 result. A small MLP ($K\rightarrow 4\rightarrow K$, 896 extra parameters total) modulates routing weights based on expert activation magnitudes. SE closes 46% of the dense gap ($33.93 \rightarrow 31.82$, target 29.36). Neither active compute scaling nor activation gating fully close the gap, motivating the structural decomposition of MoL.

D MoL Granularity Sweep

Table 9: MoL granularity sweep on WikiText-103 (50K steps). All with rank-1 MoE FFN.

Config	d_{thin}	K	Params	PPL	Speed
CT-MoE baseline	1024	—	84.7M	33.93	—
K=7 d=256 split_all	256	7	85.6M	34.83	1.51 s/s
K=7 d=256 split_mid6	256	7	85.4M	35.05	1.85 s/s
K=18 d=128	128	18	86.1M	36.67	1.08 s/s
K=40 d=64	64	40	84.7M	38.65	0.69 s/s
K=1 d=256	256	1	51.4M	39.71	4.87 s/s

$K=7$ at $d_{\text{thin}}=256$ with all layers split reaches PPL 34.83, only 0.90 worse than the baseline. Width matters more than count at iso-parameters: $d_{\text{thin}}=256$ ($K=7$) outperforms $d_{\text{thin}}=128$ ($K=18$) by 1.62 PPL. Full-width anchor layers are unnecessary: split_all (34.83) outperforms split_mid6 (35.05).

E DeltaNet Speed Crossover

Table 10: DeltaNet vs softmax attention speed (isolated block, forward+backward).

SeqLen	Softmax (ms)	DeltaNet (ms)	Ratio
512	1.07	3.12	0.34×
1024	1.89	3.18	0.59×
2048	4.04	3.33	1.21×
4096	11.06	4.81	2.30×
8192	36.08	8.93	4.04×
16384	131.81	16.88	7.81×

DeltaNet crosses softmax at $T \approx 2048$ and reaches $7.8\times$ faster at $T=16K$.

F Block-Parallel Model Sharding (Analytic)

MoL’s thin blocks have no within-layer cross-block data dependencies, which makes them a natural fit for expert-parallel placement. A prototype distributing 14 routed blocks across 3 GPUs over PCIe (with differentiable all-to-all token dispatch) yields a 38% per-GPU parameter reduction and 23% VRAM savings at a 34% wall-clock penalty; the penalty is expected to shrink substantially on NVLink. Compared to Megatron-style tensor parallelism, which incurs two all-reduces per transformer layer in the forward pass [Narayanan et al., 2021], and to standard expert-parallel MoE, where dispatch/combine collectives per MoE layer can dominate step time at scale [DeepSeek-AI, 2024], MoL’s block-parallel design requires only one all-reduce per split stage and keeps attention local to each block. A larger-scale strategy would combine block placement (primary axis), pipeline depth, and context parallelism inside long routed sub-sequences; we leave a measured throughput study to future work.

G Multi-Seed Validation

Table 11: Multi-seed validation (seeds 42, 137, 256) on WikiText-103, 50K steps.

Seed	1+3of15 (198M)	1+2of5 (87.5M)	Softmax 1+2of5 (85.3M)	Sparse 3/5 (85.3M)	Dense top-3 (85.3M)	Dense all (85.3M)
42	29.96	30.95	32.46	32.04	30.85	31.80
137	29.94	31.57	32.41	32.33	30.93	32.03
256	30.08	31.20	32.19	32.21	31.06	31.98
Mean±std	29.99 ± 0.08	31.24 ± 0.31	32.35 ± 0.15	32.19 ± 0.15	30.95 ± 0.11	31.94 ± 0.12

H Decode Latency: 3090 KV-cache Tables

The decode bench in Section 5.6 reports a single 3090 summary line; the full table is reproduced here. MoL exhibits *sub-linear* decode latency growth with context: Dense Softmax has $O(T)$ KV-cache memory traffic on every layer’s attention, while MoL splits attention across a shared softmax block (also $O(T)$) and routed DeltaNet blocks ($O(1)$ recurrent state). With 1 shared and 3 routed-active per token in 1+3of15, only $\sim 1/4$ of attention work is $O(T)$, so decode latency grows much more slowly with context than Dense.

Table 12: Decode latency per token (ms, batch=1, RTX 3090, KV cache enabled). Dense scales linearly; MoL stays flat at ~ 60 ms, dominated by a per-block Python dispatch floor that masks the smaller linear-growth term from MoL’s single shared softmax block.

Context	Dense Softmax 1.3B (ms/tok)	MoL Hybrid 2.08B (ms/tok)	Dense growth
256	6.2	63.2	—
1,024	6.9	64.2	+12%
4,096	9.6	63.3	+55%
8,192	12.9	62.5	+108%
16,384	19.6	65.2	+216%
24,576	26.5	~ 65	+328%

The asymptotic property is measured directly: *Dense’s decode latency grows $4.3\times$ over 256–24K while MoL’s grows $< 5\%$* . RTX 3090 memory caps direct measurement at ~ 24 K, so we do not extrapolate beyond. KV cache alone gives $2.74\times$ / $1.92\times$ / $6.19\times$ uplift on Dense 1.3B / Dense 0.7B / MoL at short context. The ~ 60 ms floor would shrink under a per-block-fused kernel: a partial fix (FLA-only fusion) gave +8% throughput. Full fusion is deferred. The decode bench was not re-run on H200; the prefill measurements (Section 5.6) show the same crossover regime appears on Hopper at $T \approx 96$ – 128 K (a $\sim 25\times$ shift vs 3090), so the asymptotic decode-latency story should hold on H200 starting from a higher absolute threshold.

I K_{active} Sweep at 80M (Fixed Pool $N=5$)

To isolate the effect of K_{active} at fixed pool size, we ran three additional 60K conditions on FineWeb-Edu, varying K_{active} while applying the MoL design rule $K_{\text{active}} \times d_{\text{expert}} \approx d_{\text{model}}$ and $K_{\text{active}} \times d_{\text{ff,thin}} \approx d_{\text{ff,dense}}$ (active FFN width held at ~ 4096):

Table 13: K_{active} sweep at 80M, $N=5$, FineWeb-Edu 60K steps. All conditions iso-active-FFN-width ~ 4096 . Single seed each.

Condition	K	d_{expert}	$d_{\text{ff,thin}}$	Total	Active	PPL @ 60K	Δ vs Dense 98M
1+1of5 ($K=2$)	2	512	2048	228M	119M	27.47	−2.11
1+2of5 ($K=3$)	3	320	1344	132M	98M	27.51	−2.07
1+3of5 ($K=4$, baseline)	4	256	1024	106M	95M	27.96	−1.62

$K=2$ and $K=3$ are tied within seed noise ($\sigma \approx 0.12$ from Section G); $K=4$ trails by ~ 0.5 PPL—outside seed noise but within 5σ . The direction is opposite to MoE granularity findings [Dai et al., 2024] that favour fine-grained routing. We attribute the difference to scope: at MoL, blocks include attention plus FFN (not FFN alone), so per-block compute capacity matters more than block count for shaping representations. The result is a single-seed sweep at one scale; we do not claim it generalises beyond the 80M / $N=5$ regime.

J Iso-total Gap at 1.3B: Candidate Causes

The 3.01 PPL gap between Dense Softmax 1.3B and MoL 2.08B-total at 20K steps does not reproduce the Cosmopedia-80M crossover. Three candidate causes are consistent with the missed crossover, but cannot be separated from a single trio of runs:

Active per-layer FFN capacity. At 1.3B, MoL’s active capacity (4 active blocks at $d_{\text{ff,thin}}=2048$) totals 8192, matched to Dense’s $d_{\text{ff}}=8192$. On Cosmopedia 80M, by contrast, MoL’s active capacity ($3 \times 1024 = 3072$) was $2.74\times$ Dense’s $d_{\text{ff}}=1120$, leaving a much larger margin to amortise routing overhead.

Within-MoL sparsity. Higher at 1.3B: 4-of-15 active vs 3-of-5 on Cosmopedia 1+2of5. More routed pool to specialise across at the same active count.

Total-parameter undertraining. On a total-parameter basis, MoL sees only 9.6 tokens/param at 20B tokens, well below Chinchilla-optimal ($\sim 20\times$) for a 2.08B-total model. Dense 1.3B sees 15.4 tokens/param.

A stronger crossover claim at 1.3B would require a multi-seed budget sweep we did not run; whether the gap is structural, undertraining, or optimizer/router/aux-loss tuning specific to this scale is not separable here.

K Zero-shot Evaluation: Per-Task Details

Run config: lm-eval-harness 0.4.7, bf16, batch size 64 (47.5 GB peak on H100 NVL), max length 2048, GPT-2 tokenizer (matches training). Each model is loaded via a small `lm_eval.api.model.LM` subclass that reads the architecture cfg from the checkpoint and supports batched length-bucketed loglikelihood. MMLU defaults to 5-shot per its task config; all other benchmarks are 0-shot.

Table 14 extends the headline table (Table 6) with TruthfulQA mc1, raw `acc` for HellaSwag/ARC/PIQA (whose headline metric is `acc_norm`), and stderr per cell.

Table 14: Zero-shot accuracy with stderr (%). Stderr from lm-eval-harness 1000-iter bootstrap.

Task	Metric	Dense 1.3B	Dense 0.7B	MoL 0.61B/2.08B
MMLU 5-shot	acc	24.31 $\pm$ 0.4	23.98 $\pm$ 0.4	23.07 $\pm$ 0.4
HellaSwag	acc	33.49 $\pm$ 0.5	26.81 $\pm$ 0.4	27.51 $\pm$ 0.4
HellaSwag	acc_norm	39.71 $\pm$ 0.5	34.10 $\pm$ 0.5	35.10 $\pm$ 0.5
ARC-Easy	acc	59.05 $\pm$ 1.0	54.04 $\pm$ 1.0	54.55 $\pm$ 1.0
ARC-Easy	acc_norm	53.07 $\pm$ 1.0	48.86 $\pm$ 1.0	48.15 $\pm$ 1.0
ARC-Challenge	acc	24.74 $\pm$ 1.3	24.49 $\pm$ 1.3	24.83 $\pm$ 1.3
ARC-Challenge	acc_norm	26.88 $\pm$ 1.3	25.43 $\pm$ 1.3	26.02 $\pm$ 1.3
PIQA	acc	65.94 $\pm$ 1.1	62.79 $\pm$ 1.1	63.49 $\pm$ 1.1
PIQA	acc_norm	65.56 $\pm$ 1.1	63.98 $\pm$ 1.1	64.09 $\pm$ 1.1
WinoGrande	acc	51.62 $\pm$ 1.4	49.33 $\pm$ 1.4	54.38 $\pm$ 1.4
TruthfulQA mc1	acc	21.18 $\pm$ 1.4	23.13 $\pm$ 1.5	21.91 $\pm$ 1.4
TruthfulQA mc2	acc	37.83 $\pm$ 1.4	40.89 $\pm$ 1.5	39.55 $\pm$ 1.4

Two patterns are worth noting beyond the main-body summary:

Inverse scaling on TruthfulQA. Dense 0.7B leads on both mc1 and mc2 — a documented pattern at small scale where the smaller model knows fewer plausible-but-false alternatives, shrinking its hallucination surface [Lin et al., 2022]. MoL sits between the two dense baselines, consistent with its capacity placement.

WinoGrande. MoL 0.61B-active wins this coreference task by 2.76 pts vs Dense 1.3B and 5.05 pts vs iso-active Dense 0.7B. WinoGrande is the only task where additional total capacity in MoL (2.08B vs 1.31B) appears to help despite sparse activation; the per-layer mixture of 14 specialised routed blocks may better match the heterogeneous coreference contexts than a single wide FFN.